

Problem 1

Given the Euler angle set {Roll (ψ), Pitch (θ), Yaw (ϕ)}: $\text{Rot}_z(\phi) \text{Rot}_y(\theta) \text{Rot}_x(\psi)$:

- Derive the resulting rotation matrix in terms of ϕ , θ , and ψ .
- Provide the inverse solution. In other words, given a rotation matrix $\mathbf{R} = [r_{ij}]$, determine the formulas for ϕ , θ , and ψ in terms of the entries in $\mathbf{R}$.
- At what value of which angle ϕ , θ , ψ does the inverse solution degenerate? What happens to the remaining two angles?
- Using a sketch of the intermediate rotations to assist, describe why this is a degeneracy.

Part (a): Forward Rotation Matrix

The three elementary rotations are

$$\text{Rot}_z(\phi) = \begin{bmatrix} c\phi & -s\phi & 0 \\ s\phi & c\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \text{Rot}_y(\theta) = \begin{bmatrix} c\theta & 0 & s\theta \\ 0 & 1 & 0 \\ -s\theta & 0 & c\theta \end{bmatrix}, \quad \text{Rot}_x(\psi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\psi & -s\psi \\ 0 & s\psi & c\psi \end{bmatrix}.$$

Multiplying the right-hand pair first,

$$\text{Rot}_y(\theta) \text{Rot}_x(\psi) = \begin{bmatrix} c\theta & s\theta s\psi & s\theta c\psi \\ 0 & c\psi & -s\psi \\ -s\theta & c\theta s\psi & c\theta c\psi \end{bmatrix},$$

and then pre-multiplying by $\text{Rot}_z(\phi)$:

$$\mathbf{R}(\psi, \theta, \phi) = \begin{bmatrix} c\phi c\theta & c\phi s\theta s\psi - s\phi c\psi & c\phi s\theta c\psi + s\phi s\psi \\ s\phi c\theta & s\phi s\theta s\psi + c\phi c\psi & s\phi s\theta c\psi - c\phi s\psi \\ -s\theta & c\theta s\psi & c\theta c\psi \end{bmatrix}$$

Part (b): Inverse Solution

From the third row and first column of the result above,

$$s\theta = -r_{31}, \quad c\theta = \pm \sqrt{r_{11}^2 + r_{21}^2} \quad \text{since} \quad r_{11}^2 + r_{21}^2 = c\phi^2 c\theta^2 + s\phi^2 c\theta^2 = c\theta^2.$$

With $c\theta$ known, the first column gives ϕ and the third row gives ψ :

$$\theta = \text{atan2}\left(-r_{31}, \pm \sqrt{r_{11}^2 + r_{21}^2}\right)$$

$$\phi = \text{atan2}\left(\frac{r_{21}}{c\theta_{\pm}}, \frac{r_{11}}{c\theta_{\pm}}\right), \quad \psi = \text{atan2}\left(\frac{r_{32}}{c\theta_{\pm}}, \frac{r_{33}}{c\theta_{\pm}}\right)$$

The $\pm$ on the square root gives the two standard solution branches.

Part (c): Degeneracy

The solution degenerates at $\theta = \frac{\pi}{2} + \pi n$ (a 90° pitch).

At that pitch $c\theta = 0$, so $r_{11} = r_{21} = r_{32} = r_{33} = 0$ and both the ϕ and ψ expressions become $\text{atan2}(0/0, 0/0)$. It becomes impossible to find their individual values, only their sum or difference.

Part (d): Why It Is a Degeneracy

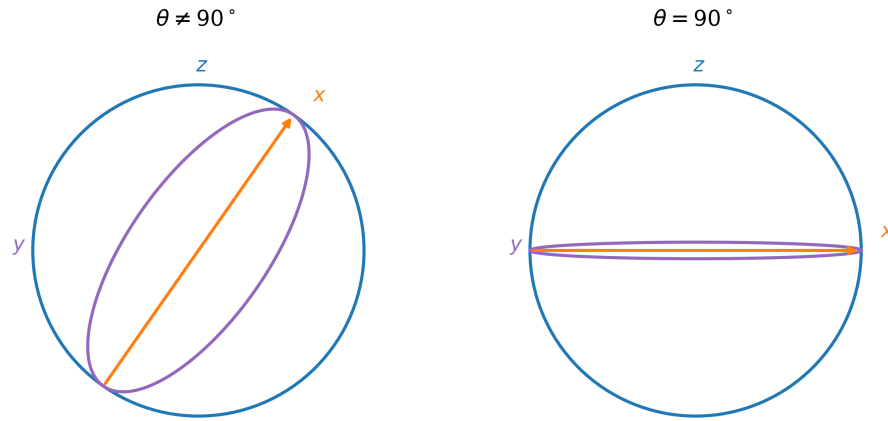


Figure 1: Intermediate rotations. At left ($\theta \neq 90^\circ$) the roll axis $\hat{x}$ and the yaw axis $\hat{z}$ are independent. At right ($\theta = 90^\circ$) the roll axis has been pitched into alignment with the yaw axis.

When you rotate pitch to 90° , the roll axis becomes parallel with the yaw axis, so any rotation from then on spins both. The two rotations are no longer independent: only the combined angle acts on the body, which is why the inverse solution can recover the sum/difference but not ϕ and ψ separately. Substituting $\theta = \pi/2$ into part (a) gives,

$$\mathbf{R}(\psi, \frac{\pi}{2}, \phi) = \begin{bmatrix} 0 & s(\psi - \phi) & c(\psi - \phi) \\ 0 & c(\psi - \phi) & -s(\psi - \phi) \\ -1 & 0 & 0 \end{bmatrix},$$

which depends on ϕ and ψ only through the difference $\psi - \phi$.

Problem 2

Given the following rotation matrix,

$$\mathbf{R} = \begin{bmatrix} 0.6533 & 0.6403 & 0.4040 \\ -0.3488 & 0.7282 & -0.5900 \\ -0.6719 & 0.2445 & 0.6991 \end{bmatrix},$$

- Prove that it is a rotation matrix.
- What are the ZYZ angles for this rotation matrix?
- What is the angle and axis representation for this rotation matrix?
- What is the Quaternion representation?

Part (a): Proof That $\mathbf{R}$ Is a Rotation

A matrix is a rotation if it is orthonormal with a determinant of +1. Evaluating both conditions:

$$\det(\mathbf{R}) = 1, \quad \mathbf{R}\mathbf{R}^T \approx \mathbf{I}$$

(Evaluated by calculator; the residual in $\mathbf{R}\mathbf{R}^T - \mathbf{I}$ is only round-off from the four-digit entries.) Since the columns are mutually orthogonal unit vectors and the determinant is +1 rather than -1, $\mathbf{R}$ is a proper rotation.

Part (b): ZYZ Euler Angles

For $\mathbf{R} = \text{Rot}_z(\phi) \text{Rot}_y(\theta) \text{Rot}_z(\psi)$:

$$\theta_{\pm} = \text{atan2}\left(\pm\sqrt{r_{13}^2 + r_{23}^2}, r_{33}\right), \quad \phi = \text{atan2}\left(\frac{r_{23}}{s\theta_{\pm}}, \frac{r_{13}}{s\theta_{\pm}}\right), \quad \psi = \text{atan2}\left(\frac{r_{32}}{s\theta_{\pm}}, \frac{-r_{31}}{s\theta_{\pm}}\right)$$

Taking the positive branch:

$$\theta = 45.65^\circ, \quad \phi = -55.60^\circ, \quad \psi = 20.00^\circ$$

Part (c): Angle–Axis Representation

$$\theta = \text{atan2}\left(\frac{1}{2} \left\| \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} \right\|_2, \frac{\text{tr}(\mathbf{R}) - 1}{2}\right), \quad \hat{\mathbf{k}} = \frac{1}{2s\theta} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix}$$

$$\theta = 57.30^\circ, \quad \hat{\mathbf{k}} = \begin{bmatrix} 0.496 \\ 0.639 \\ -0.588 \end{bmatrix}$$

Part (d): Quaternion Representation

$$q_0 = \pm \sqrt{\frac{1 + \text{tr}(\mathbf{R})}{4}}, \quad q_1 = \frac{r_{32} - r_{23}}{4q_0}, \quad q_2 = \frac{r_{13} - r_{31}}{4q_0}, \quad q_3 = \frac{r_{21} - r_{12}}{4q_0}$$

$$Q = \left[0.878, \begin{bmatrix} 0.238 & 0.307 & -0.282 \end{bmatrix}^T \right]$$

Problem 3

What is the rotation matrix and angle-axis vector ($\mathbf{\Omega} = \hat{\mathbf{k}}\theta$) that are described by the following quaternions:

- (a) $[1, [0 \ 0 \ 0]^T]$
- (b) $[0, [0 \ 0 \ 1]^T]$
- (c) $[0, [1 \ 0 \ 0]^T]$
- (d) $[0.5332, [0.5928 \ 0.08311 \ 0.5978]^T]$

Each quaternion $[q_0, [q_1 \ q_2 \ q_3]^T]$ maps to a rotation by

$$\mathbf{R} = \begin{bmatrix} 1 - 2(q_2^2 + q_3^2) & 2(q_1q_2 - q_0q_3) & 2(q_1q_3 + q_0q_2) \\ 2(q_1q_2 + q_0q_3) & 1 - 2(q_1^2 + q_3^2) & 2(q_2q_3 - q_0q_1) \\ 2(q_1q_3 - q_0q_2) & 2(q_2q_3 + q_0q_1) & 1 - 2(q_1^2 + q_2^2) \end{bmatrix},$$

$$\theta = 2 \arccos(q_0), \quad \hat{\mathbf{k}} = \frac{1}{\sin(\theta/2)} \begin{bmatrix} q_1 \\ q_2 \\ q_3 \end{bmatrix}.$$

Part (a): $[1, [0 \ 0 \ 0]^T]$

$$\mathbf{R} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \theta = 0^\circ, \quad \hat{\mathbf{k}} = \vec{0}$$

This is the identity quaternion: there is no rotation, so the axis is arbitrary (taken as $\vec{0}$) and $\mathbf{\Omega} = \hat{\mathbf{k}}\theta = \vec{0}$.

Part (b): $[0, [0 \ 0 \ 1]^T]$

$$\mathbf{R} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \theta = 180^\circ, \quad \hat{\mathbf{k}} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Part (c): $[0, [1 \ 0 \ 0]^T]$

$$\mathbf{R} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad \theta = 180^\circ, \quad \hat{\mathbf{k}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Part (d): $[0.5332, [0.5928 \ 0.08311 \ 0.5978]^T]$

$$\mathbf{R} = \begin{bmatrix} 0.271 & -0.539 & 0.797 \\ 0.736 & -0.418 & -0.533 \\ 0.620 & 0.732 & 0.283 \end{bmatrix}, \quad \theta = 115.56^\circ, \quad \hat{\mathbf{k}} = \begin{bmatrix} 0.701 \\ 0.098 \\ 0.707 \end{bmatrix}$$

Problem 4

An object is spinning with some constant angular velocity ${}^0\boldsymbol{\omega}$ such that after time t it has rotated $\theta = \|\boldsymbol{\omega}\|_2 t$ radians about ${}^0\hat{\mathbf{k}} = {}^0\boldsymbol{\omega} / \|\boldsymbol{\omega}\|_2$. Its orientation is then ${}^0\mathbf{R}_{obj} = {}^0\mathbf{R}_i(\theta, \hat{\mathbf{k}}) {}^i\mathbf{R}_{obj}$. Using the angle-axis representation, derive $\dot{\mathbf{R}} = d\mathbf{R}/dt$, which by the chain rule yields

$$\frac{d\mathbf{R}}{dt} = \left(\frac{d}{d\theta} {}^0\mathbf{R}_i(\theta, \hat{\mathbf{k}}) {}^i\mathbf{R}_{obj} \right) \dot{\theta} + \left(\frac{d}{d\hat{\mathbf{k}}} {}^0\mathbf{R}_i(\theta, \hat{\mathbf{k}}) {}^i\mathbf{R}_{obj} \right) \dot{\hat{\mathbf{k}}},$$

and evaluate at $\theta = t = 0$, before the orientation is modified by rotation.

Setup: Rodrigues' Formula

From (3.41) in Hollerbach's notes,

$$\mathbf{R}_{\hat{\mathbf{k}}}(\theta) = \mathbf{I} c\theta + [\hat{\mathbf{k}}]_{\times} s\theta + \hat{\mathbf{k}}\hat{\mathbf{k}}^T(1 - c\theta).$$

The Two Chain-Rule Terms

$$\frac{d\mathbf{R}}{dt} = \left(\frac{d}{d\theta} {}^0\mathbf{R}_i(\theta, \hat{\mathbf{k}}) {}^i\mathbf{R}_{obj} \right) \dot{\theta} + \left(\frac{d}{d\hat{\mathbf{k}}} {}^0\mathbf{R}_i(\theta, \hat{\mathbf{k}}) {}^i\mathbf{R}_{obj} \right) \dot{\hat{\mathbf{k}}}$$

The angular velocity is constant, so

$$\hat{\mathbf{k}} = \frac{{}^0\boldsymbol{\omega}}{\|\boldsymbol{\omega}\|_2}, \quad \frac{d\boldsymbol{\omega}}{dt} = 0 \quad \therefore \quad \dot{\hat{\mathbf{k}}} = 0,$$

which gets rid of the second term. For the first term, $\theta = \|\boldsymbol{\omega}\|_2 t$ gives

$$\dot{\theta} = \frac{d\theta}{dt} = \|\boldsymbol{\omega}\|_2.$$

Differentiating Rodrigues' Formula

$$\frac{d}{d\theta} \left(\mathbf{R}_{\hat{\mathbf{k}}}(\theta) \right) = -\mathbf{I} s\theta + [\hat{\mathbf{k}}]_{\times} c\theta + \hat{\mathbf{k}}\hat{\mathbf{k}}^T s\theta \quad \therefore \quad \left. \frac{d}{d\theta} \left(\mathbf{R}_{\hat{\mathbf{k}}}(\theta) \right) \right|_{\theta=0} = [\hat{\mathbf{k}}]_{\times}.$$

Evaluating at $\theta = t = 0$

Substituting both results,

$$\frac{d\mathbf{R}}{dt} = \left(\underbrace{\frac{d}{d\theta} {}^0\mathbf{R}_i(\theta, \hat{\mathbf{k}})}_{[\hat{\mathbf{k}}]_{\times}} {}^i\mathbf{R}_{obj} \right) \|{}^0\boldsymbol{\omega}\|_2 = [\hat{\mathbf{k}}]_{\times} {}^i\mathbf{R}_{obj} \|{}^0\boldsymbol{\omega}\|_2.$$

At $\theta = 0$ the incremental rotation is the identity, so the orientation is unmodified:

$${}^0\mathbf{R}_{obj}(0) = {}^0\mathbf{R}_i(0, \hat{\mathbf{k}}) {}^i\mathbf{R}_{obj} = \mathbf{I} \cdot {}^i\mathbf{R}_{obj} \quad \therefore \quad {}^0\mathbf{R}_{obj}(0) = {}^i\mathbf{R}_{obj}.$$

Finally, the scalar $\|{}^0\boldsymbol{\omega}\|_2$ gets multiplied inside the skew operator:

$$\|{}^0\boldsymbol{\omega}\|_2 [\hat{\mathbf{k}}]_{\times} = \|{}^0\boldsymbol{\omega}\|_2 \left[\frac{\boldsymbol{\omega}}{\|{}^0\boldsymbol{\omega}\|_2} \right]_{\times} = [\boldsymbol{\omega}]_{\times}.$$

$$\frac{d\mathbf{R}}{dt} = [\boldsymbol{\omega}]_{\times} {}^0\mathbf{R}_{obj}$$

Problem 5

Consider the combination of robot, square table, block, and camera below, with associated coordinate systems as shown. The relative locations of robot, table, block, and camera are shown in Fig. 2. The block is located at the center of the table.

- (a) Find 0T_1 , 1T_2 , 2T_3 , and 0T_3 by inspection.
- (b) Assume the block has moved. It is tracked by the camera and is known to exist at location ${}^3d_{32} = [-0.5 \ 0.5 \ 3]^T$. Where is the block in the robot's coordinate system, i.e. what is ${}^0d_{02}$?

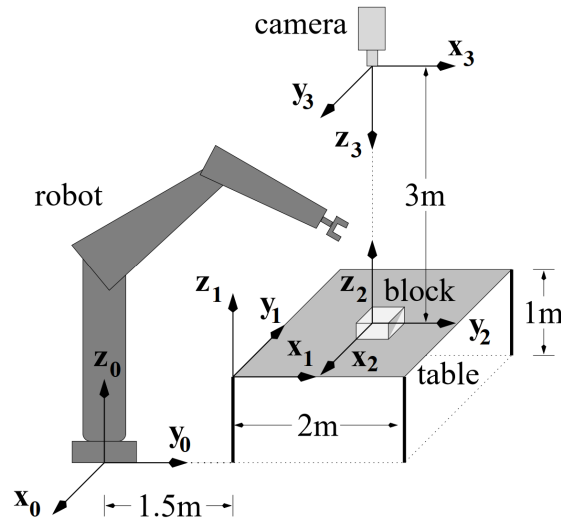


Figure 2: Robot, square table, block, and camera with their associated coordinate systems.

Part (a): Transforms by Inspection

$${}^0T_1 = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 1.5 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^1T_2 = \begin{bmatrix} 0 & 1 & 0 & 1 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad {}^2T_3 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Chaining them,

$${}^0T_3 = {}^0T_1 {}^1T_2 {}^2T_3$$

$${}^0T_3 = \begin{bmatrix} 0 & 1 & 0 & -1 \\ 1 & 0 & 0 & 2.5 \\ 0 & 0 & -1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Part (b): Block Location in the Robot Frame

Given ${}^3\mathbf{d}_{32} = [-0.5 \quad 0.5 \quad 3]^T$, map it through ${}^0\mathbf{T}_3$ in homogeneous coordinates:

$$\begin{bmatrix} {}^0\mathbf{d}_{02} \\ 1 \end{bmatrix} = {}^0\mathbf{T}_3 \begin{bmatrix} {}^3\mathbf{d}_{32} \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & -1 \\ 1 & 0 & 0 & 2.5 \\ 0 & 0 & -1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -0.5 \\ 0.5 \\ 3 \\ 1 \end{bmatrix}$$

$${}^0\mathbf{d}_{02} = \begin{bmatrix} -0.5 \\ 2 \\ 1 \end{bmatrix}$$

Appendix A: Python Code

Problem 1: Gimbal Degeneracy Sketch

Listing 1: Problem 1.py: generates the intermediate-rotation sketch in Fig. 1.

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 from matplotlib.patches import Circle
4
5
6 def ellipse(ax, semi_major, semi_minor, tilt_deg, color):
7     """Draw the ring swept by the roll axis as an ellipse (a circle seen
8         edge-on)."""
9     t = np.linspace(0, 2 * np.pi, 400)
10    x = semi_major * np.cos(t)
11    y = semi_minor * np.sin(t)
12    a = np.radians(tilt_deg)
13    xr = x * np.cos(a) - y * np.sin(a)
14    yr = x * np.sin(a) + y * np.cos(a)
15    ax.plot(xr, yr, color=color, lw=2)
16
17 def gimbal(ax, pitch_deg, title):
18     # Outer sphere / yaw frame
19     ax.add_patch(Circle((0, 0), 1.0, fill=False, ec="tab:blue", lw=2))
20     ax.text(0.02, 1.08, r"$z$", color="tab:blue", fontsize=13, ha="center")
21
22     # Roll ring: flattens out and lines up with the x arrow as pitch -> 90
23     # deg
24     tilt = 90.0 - pitch_deg
25     ellipse(ax, 1.0, 0.45 * np.cos(np.radians(pitch_deg)) + 0.05, tilt, "
26         tab:purple")
27     ax.text(-1.12, 0.03, r"$y$", color="tab:purple", fontsize=13, va="
28         center")
29
30     # x axis after the pitch rotation
31     a = np.radians(tilt)
32     ax.annotate("", xy=(np.cos(a), np.sin(a)), xytext=(-np.cos(a), -np.sin
33         (a)),
34         arrowprops=dict(arrowstyle="-|>", color="tab:orange", lw
35             =2))
36     ax.text(np.cos(a) + 0.12, np.sin(a) + 0.08, r"$x$", color="tab:orange"
37         , fontsize=13)
38
39     ax.set_title(title, fontsize=14)
40     ax.set_xlim(-1.45, 1.45)
41     ax.set_ylim(-1.3, 1.3)
42     ax.set_aspect("equal")
43     ax.axis("off")
44
45 fig, axes = plt.subplots(1, 2, figsize=(9, 4.5))

```

```
41 gimbal(axes[0], 35.0, r"$\theta \neq 90^\circ$")
42 gimbal(axes[1], 90.0, r"$\theta = 90^\circ$")
43 fig.tight_layout()
44 fig.savefig("../Latex/Figures/Problem1D.png", dpi=300, bbox_inches="tight")
    )
```

Problem 2: ZYZ Angles, Angle–Axis, and Quaternion

Listing 2: Problem 2.py: evaluates the ZYZ angles, angle-axis pair, and quaternion for Problem 2.

```

1 import numpy as np
2 from numpy.linalg import norm
3
4
5
6 # Given rotation matrix
7 R = np.array([
8     [ 0.6533,  0.6403,  0.4040],
9     [-0.3488,  0.7282, -0.5900],
10    [-0.6719,  0.2445,  0.6991]
11 ])
12
13 r13, r23, r33 = R[0, 2], R[1, 2], R[2, 2]
14 r31, r32 = R[2, 0], R[2, 1]
15 r11, r22 = R[0, 0], R[1, 1]
16 r21, r12 = R[1, 0], R[0, 1]
17
18
19 # ZYZ Euler angles: R = Rz(phi) * Ry(theta) * Rz(psi)
20 theta = np.arctan2(np.sqrt(r13**2 + r23**2), r33)
21 phi = np.arctan2(r23, r13)
22 psi = np.arctan2(r32, -r31)
23
24 # Angle-Axis
25 diffVector = np.array([
26     [r32 - r23],
27     [r13 - r31],
28     [r21 - r12]
29 ])
30
31 theta2 = np.arctan2(.5 * norm(diffVector), (np.trace(R) - 1)/2)
32 k = 1/ (2 * np.sin(theta2)) * diffVector
33
34 # Quaternion
35
36 q_0 = np.sqrt((1 + np.trace(R))/4)
37 q_1 = (r32 - r23) / (4 * q_0)
38 q_2 = (r13 - r31) / (4 * q_0)
39 q_3 = (r21 - r12) / (4 * q_0)
40
41 print(f"phi    = {np.degrees(phi):.4f} deg")
42 print(f"theta  = {np.degrees(theta):.4f} deg")
43 print(f"psi    = {np.degrees(psi):.4f} deg")
44 print(f"theta2 = {np.degrees(theta2):.4f} deg")
45 print(f"k = {k.flatten()}")
46 print(f"q = [{q_0:.4f}, {q_1:.4f}, {q_2:.4f}, {q_3:.4f}]")

```

Problem 3: Quaternion to Rotation Matrix and Angle–Axis

Listing 3: Problem 3.py: converts each quaternion in Problem 3 to a rotation matrix and angle-axis pair.

```

1 import numpy as np
2
3 def quat_to_R_and_axis(q0, q1, q2, q3):
4     R = np.array([
5         [1 - 2*(q2**2 + q3**2),      2*(q1*q2 - q0*q3),      2*(q1*q3 + q0*
6             q2)],
7         [ 2*(q1*q2 + q0*q3), 1 - 2*(q1**2 + q3**2),      2*(q2*q3 - q0*
8             q1)],
9         [ 2*(q1*q3 - q0*q2),      2*(q2*q3 + q0*q1), 1 - 2*(q1**2 + q2
10             **2)]
11     ])
12
13     theta = 2 * np.arccos(q0)
14     s = np.sin(theta / 2)
15     if s < 1e-8:
16         k = np.array([0.0, 0.0, 0.0])
17     else:
18         k = np.array([q1, q2, q3]) / s
19
20     return R, theta, k
21
22 quaternions = {
23     "a": (1, 0, 0, 0),
24     "b": (0, 0, 0, 1),
25     "c": (0, 1, 0, 0),
26     "d": (0.5332, 0.5928, 0.08311, 0.5978),
27 }
28
29 for label, q in quaternions.items():
30     R, theta, k = quat_to_R_and_axis(*q)
31     print(f"--- ({label}) q = {q} ---")
32     print(f"R =\n{R}")
33     print(f"theta = {np.degrees(theta):.4f} deg")
34     print(f"k = {k}")
35     print()

```

Appendix B: Hand Calculations

Corrections made when typing up the hand calcs below:

1. **Problem 1(b), page 1:** the boxed yaw formula was written as $\phi = \text{atan2}\left(\frac{r_{21}}{c\theta_{\pm}}, \frac{r_{12}}{c\theta_{\pm}}\right)$, but r_{12} doesn't show up in the ϕ equation. The correct value comes from the first column, $r_{21} = s\phi c\theta$ and $r_{11} = c\phi c\theta$ (the same terms $c\theta = \sqrt{r_{11}^2 + r_{21}^2}$ above it is built from).
2. **Problem 3(d), page 6:** the (1,2) entry of the rotation matrix was written as -0.639 . Evaluating $2(q_1q_2 - q_0q_3) = 2((0.5928)(0.08311) - (0.5332)(0.5978))$ gives -0.539 , matching the neighbouring entries $(0.271, 0.797, 0.736, -0.418, -0.533, 0.620, 0.732, 0.283)$ and the boxed axis $\hat{\mathbf{k}} = [0.701 \ 0.098 \ 0.707]^T$ exactly.

Problem 1

$$\text{Rot } Z(\phi) \text{ Rot } Y(\theta) \text{ Rot } X(\psi)$$

$$\begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \psi & -\sin \psi \\ 0 & \sin \psi & \cos \psi \end{bmatrix}$$

$$\begin{bmatrix} \cos \theta & \sin \theta \sin \psi & \sin \theta \cos \psi \\ 0 & \cos \psi & -\sin \psi \\ -\sin \theta & \cos \theta \sin \psi & \cos \theta \cos \psi \end{bmatrix}$$

a) $R(\psi, \theta, \phi):$

$$\begin{bmatrix} \cos \phi \cos \theta & \cos \phi \sin \theta \sin \psi - \sin \phi \cos \psi & \cos \phi \sin \theta \cos \psi + \sin \phi \sin \psi \\ \sin \phi \cos \theta & \sin \phi \sin \theta \sin \psi + \cos \phi \cos \psi & \sin \phi \sin \theta \cos \psi - \cos \phi \sin \psi \\ -\sin \theta & \cos \theta \sin \psi & \cos \theta \cos \psi \end{bmatrix}$$

b) $\sin \theta = -r_{31}$

$$\cos \theta = \sqrt{r_{11}^2 + r_{21}^2}$$

$$\sqrt{r_{11}^2 + r_{21}^2} = \sqrt{\cos^2 \theta \cos^2 \psi + \sin^2 \theta \cos^2 \psi}$$

$$\therefore \theta = \arctan 2(-r_{31}, \pm \sqrt{r_{11}^2 + r_{21}^2})$$

$$\phi = \arctan 2\left(\frac{r_{21}}{\cos \theta_{\pm}}, \frac{r_{12}}{\cos \theta_{\pm}}\right)$$

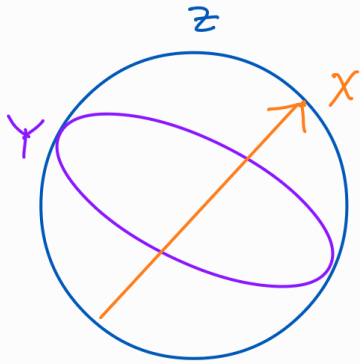
$$\psi = \arctan 2\left(\frac{r_{32}}{\cos \theta_{\pm}}, \frac{r_{33}}{\cos \theta_{\pm}}\right)$$

C) $\theta = \pi/2 + \pi n$ (90 degree Pitch)

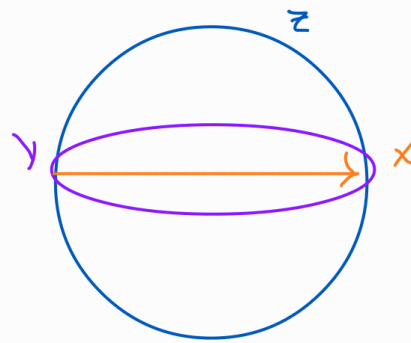
It becomes impossible to find their individual value, only the sum/diff.

D)

$\theta \neq 90$



$\theta = 90$



When you rotate pitch to 90° , it becomes parallel with the roll axis, so any rotation from then on spins both.

Problem 2

$$R = \begin{bmatrix} .6533 & .6403 & .4040 \\ -.3488 & .7282 & -.5900 \\ -.6719 & .2445 & .6991 \end{bmatrix}$$

a) $\det(R) = 1$ I used a calculator ☺

$$R \cdot R^T \approx \underline{I}$$

b) for zyz :

$$\theta_{\pm} = \text{atan2}(\pm \sqrt{r_{13}^2 + r_{23}^2}, r_{33})$$

$$\phi = \text{atan2}\left(\frac{r_{23}}{\sin \theta_{\pm}}, \frac{r_{13}}{\sin \theta_{\pm}}\right)$$

$$\psi = \text{atan2}\left(\frac{r_{32}}{\sin \theta_{\pm}}, -\frac{r_{31}}{\sin \theta_{\pm}}\right)$$

$$\theta = 45.65^\circ \quad \phi = -55.6^\circ \quad \psi = 20.0^\circ$$

c) $\theta = \text{atan2}\left(\frac{1}{2} \left\| \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} \right\|_2, \frac{\text{tr}(R) - 1}{2}\right)$

$$\hat{k} = \frac{1}{2 \sin \theta} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix}$$

$$\therefore \theta = 57.3^\circ$$

$$\hat{k} = \begin{bmatrix} .496 \\ .639 \\ -.588 \end{bmatrix}$$

$$d) \quad q_0 = \pm \sqrt{\frac{1 + \text{tr}(R)}{4}}$$

$$q_1 = \frac{r_{32} - r_{23}}{4q_0}$$

$$q_2 = \frac{r_{13} - r_{31}}{4q_0}$$

$$q_3 = \frac{r_{21} - r_{12}}{4q_0}$$

$$Q = [878, [0.238, 0.307, -0.282]^T]$$

Problem 3

a) $R([1, [0, 0, 0]^T]) =$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\theta = 0^\circ$$

$$\hat{k} = \vec{0}$$

b) $R([0, [0, 0, 1]^T]) =$

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\theta = 180^\circ$$

$$\hat{k} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$C) R([0, [1, 0, 0]^T]) =$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad \theta = 180^\circ$$

$$\hat{k} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$D) R([.5332, [.5928, .08311, .5978]^T]) =$$

$$\begin{bmatrix} .271 & -.539 & .797 \\ .736 & -.418 & -.533 \\ .620 & .732 & .283 \end{bmatrix} \quad \theta = 115^\circ$$

$$\hat{k} = \begin{bmatrix} .701 \\ .098 \\ .707 \end{bmatrix}$$

problem 4

$$R_K(\theta) = \underline{I} \cos \theta + [K]_x \sin \theta + K K^T (1 - \cos \theta)$$

$$\frac{dR}{dt} = \left(\frac{d}{d\theta} {}^0R_i(\theta, \hat{K})^i R_{obj} \right) \dot{\theta} + \left(\frac{d}{d\hat{K}} {}^0R_i(\theta, \hat{K})^i R_{obj} \right) \dot{\hat{K}}$$

$$\hat{K} = \frac{{}^0w}{\|w\|_2}, \quad \frac{dw}{dt} = 0 \therefore \dot{\hat{K}} = 0$$

$$\theta = \|w\|_2 t \quad \therefore \frac{d\theta}{dt} = \|w\|_2$$

From $R_K(\theta)$

$$\frac{d}{d\theta} (R_K(\theta)) =$$

$$-\underline{I} S\theta + [K]_x C\theta + K K^T S\theta$$

$$\left. \frac{d}{d\theta} (R_K(\theta)) \right|_{\theta=0} = [K]_x$$

$$\frac{dR}{dt} = \left(\frac{d}{d\theta} {}^0R_i(\theta, K) {}^iR_{obj} \right) \| {}^0w \|_2$$

$$[K]_x = [K_x] \cdot {}^0R_{obj} \cdot \| {}^0w \|_2$$

$$\| {}^0w \|_2 \cdot [K_x] \cdot {}^0R_{obj}(0) = {}^0R_i(0, \hat{K}) {}^iR_{obj}$$

$$\| {}^0w \|_2 \cdot \left[\frac{w}{\| {}^0w \|_2} \right]_x \cdot {}^0R_{obj}(0) = {}^iR_{obj}$$

$$= [w]_x$$

$$\frac{dR}{dt} = [w]_x \cdot {}^0R_{obj}$$

Problem 5

$$a) \quad {}^0T_1 = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 1.5 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1T_2 = \begin{bmatrix} 0 & 1 & 0 & 1 \\ -1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2T_3 = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0T_3 = {}^0T_1 {}^1T_2 {}^2T_3 =$$

$$\begin{bmatrix} 0 & 1 & 0 & -1 \\ 1 & 0 & 0 & 2.5 \\ 0 & 0 & -1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$b) \quad {}^3d_{32} = [-.5, .5, 3]^T$$

$$\begin{bmatrix} 0 \\ \vec{d}_{02} \\ 1 \end{bmatrix} = {}^0T_3 \cdot \begin{bmatrix} \vec{{}^3d_{32}} \\ 1 \end{bmatrix}$$

$${}^0d_{02} = \begin{bmatrix} -.5 \\ 2 \\ 1 \end{bmatrix}$$